

Network Neighbors: A small-world analysis on a network of Twitch

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Abstract

Small-world network analysis. This principle—the small-world phenomenon—structures various types of network connections, including social networks, collaborative networks, communication networks, and information networks. The result is a network structure characterized by short path lengths between nodes and a high degree of clustering. This means that, whether in social media or other contexts, small-world networks exhibit highly efficient dissemination capabilities and tightly knit community structures. Consequently, people's interactive networks tend to display significant clustering effects based on geographic location, interests, and other factors.

In recent years, with the rapid development of information technology, people's connections are no longer constrained by geographical location. The rise of major live broadcasting platforms has provided a favorable environment for the small-world phenomenon in social systems. Even if two people are thousands of miles apart, they can connect at the same node due to shared interests. This creates the conditions for the formation of the small-world phenomenon in the social space. By facilitating interaction and communication between users, live streaming platforms enable the rapid dissemination of information and resources, further enhancing the characteristics of small-world networks.

Twitch is a hot streaming platform, people are keen to share their lives on it or interact with Streamer, in pop-ups, and also the Streamer can earn considerable income on it, gradually forming a complete social network, this paper aims, to find the small world analysis of the social

network in Twitch, firstly this thesis gives an introduction to what small-worldness (or the small-world phenomenon) is, followed by the methodology of the study, experiments, and the results, and finally all of the above work will be systematically discussed and summarized.

Introduction

Following the rapid development of information and technology, the information contact between people from the earliest, letter/horse transmission, to the later telephone transmission, to the modern electronic transmission, we will find that the distance is no longer the main obstacle between people to produce the main contact between people, some of you feel that you are separated from the "distant" people, in fact, and you are "very close". Some people you think are "far away" from you are actually "very close" to you.

The Small World Network represent this phenomenon (it is also known as the Small World Phenomenon), and the origins of Small Worlds can be traced back to a 1967 experiment (also known as Six Degrees of Separation) conducted by a sociologist named Stanley Milgram, a professor at Harvard University, who found that even in a large society, any two people who don't know each other often have to go through only six intermediaries to make a connection. He found that, even in a large society, any contact between two people who don't know each other often goes through only six intermediaries.

This was followed by a significant paper published in 1998 by mathematicians Duncan Watts and Steven Strogatz in the Nature journal. They detailed the small-world phenomenon systematically through mathematical models and computer simulations. They are credited with coming up with the Watts-Strogatz Model, demonstrating that various networks we encounter in real life— like social networks or neural networks — exhibit small-world properties because they have short path lengths while having high aggregation coefficients.

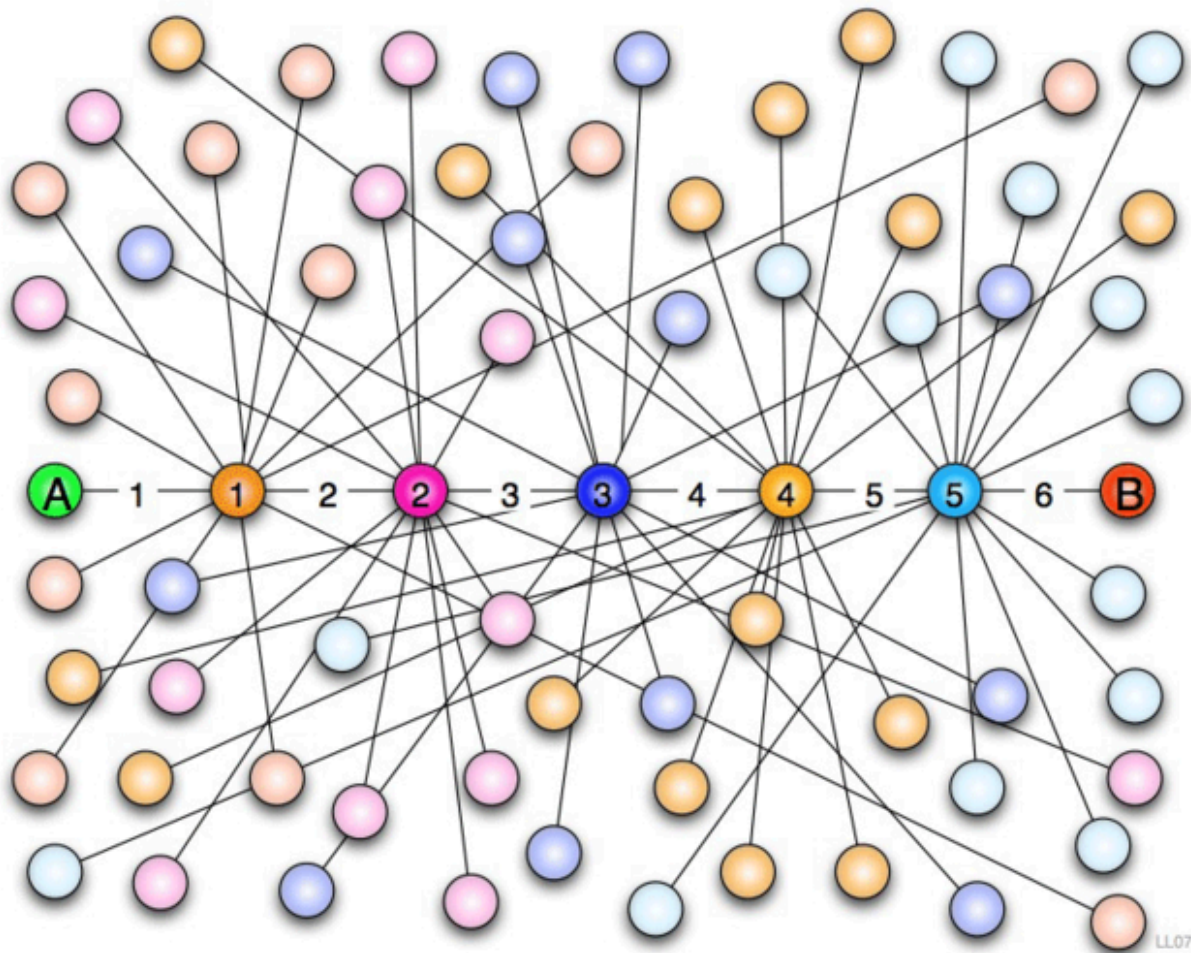


Figure 1. Small world network example

These early studies suggest that small-world patterns are prevalent in many networks in nature and society. In this paper we will investigate small-world network analysis between Twitch streamers and their followers, elucidating the small-world properties in the Twitch network through, among other things, features such as shot path lengths, and hopefully providing a clear framework for the network.

Related Works in the area

In the most recent fifty years, many fundamental social network researchers acknowledged that the small-world attributes of various networks are effectively willful using the Watts-Strogatz

model. In a series of experiments, such as the "six degrees of separation" ones conducted by Milgram and other researchers on social networks' networks, it has been shown that small-world theory is generally applicable to many social networks. Social media connection usually shows the features of strong bond, such as its being smaller, tighter, and having most members in closer proximity. Such environments confirm the existence of structural variations which are emotionally and mentally asymptomatic while the opposite two poles of the small-world syndrome are presented in a person's surroundings.

Among other topics, the effect of online social networks, particularly those of gaming networks, on the development of small-world structures has made the concept more popular. As long as the study "Twitch Gamers: a Data Set for Evaluating Proximity Preserving and Structural Role-based Node Embeddings" brought into the daylight in 2021 by Rozemberczki & Sarkar provides copious volumes of data to dig into the essence of social systems and networks; then what does Stanford have to, after all? Consequently, their work illustrates an intricate direct-connection network, which is suitable for imitating proximity-preserving and structure-role-based node embeddings. This research emphasizes the fact that the complex relational ties existing within the online gaming communities quite definingly mimic the real-life small-world constructions.

While a significant number of researches on small-world networks has investigated social networks as general case and some online platforms, around less study focuses on the structure characteristics of actual gaming communities or platforms such as Twitch. Therefore, the main goal of this study is to address this frequently omitted topic by closely examining the small-world aspects of the Twitch community while observing their influence on community formation and information transfer.

Methodology

In this project, we use the Twitch Gamers Social Network on software called SNAPS (Stanford Network Analysis Project). This data consists of the social network of Twitch users collected from a public API in the spring of 2018. The nodes are Twitch users, and the edges are the mutual following relationships between them. (*SNAP: Network Datasets: Twitch Gamers Social Network*, n.d.)

In our study, we carefully selected a subset of data through random sampling, comprising 2500 nodes and their respective edges from the larger dataset. The choice of random sampling was based on its effectiveness in analyzing small-world networks, which are characterized by short path lengths between nodes and high clustering coefficients (Wikipedia Contributors, 2023).

Although snowball sampling—a method involving iterative expansion from initial nodes—was considered, its application posed significant risks to the integrity of our small-world analysis. The potential introduction of biases could distort our findings, undermining the accuracy of our experimental outcomes (Wang et al., 2011).

Similarly, the exploration of probability sampling methods based on centrality measures or other network attributes faced challenges similar to those of snowball sampling. These methods, while potentially informative in other contexts, were deemed unsuitable for our study due to concerns about potential bias and distorted outcomes (Wang et al., 2011).

The decision to use sampling methodologies was driven by practical considerations related to computational resources. Given the computational demands of network analysis tasks, limiting the dataset size through sampling was crucial to ensure that our analyses could be completed within a reasonable timeframe (Wikipedia Contributors, 2024).

Additionally, to maintain accuracy and ensure reliable results —particularly in measurements reliant on network connectivity, such as path length calculations and clustering coefficients—we focused exclusively on the largest connected component of the network. This method helped us reduce the effect of disconnected nodes and find meaningful patterns in the data.

After conducting random sampling and identifying the largest connected component, the resulting graph contains 999 nodes and 1,550 edges.

Dataset description

The dataset comprises 168,115 nodes representing individual Twitch users. These nodes are interconnected by a total of 6,797,557 edges, indicating the relationships or interactions between users.

Each node in the dataset is characterized by several features:

- **Views:** Numerical values representing the total number of views each user has received.
- **Mature:** A boolean value indicating whether the user creates explicit content.
- **Life_time:** The date when the user's account transitioned to a state of inactivity or "dead" status.
- **Created_at:** The date when the user's account was originally created.
- **Updated_at:** The date of the most recent update to the user's account information.
- **Dead_account:** A boolean value indicating whether the user's account is currently active or inactive.
- **Language:** A categorical value denoting the language in which the user's content is primarily created.
- **Affiliate:** An indicator of the user's Twitch affiliate status, reflecting whether the account has generated revenue from its content.

However, for our analysis, we did not utilize any of the features, as our primary focus was solely on determining whether the network exhibits small-world characteristics.

Research Methodology used

Analysis

We decided to conduct a small-world analysis of the network and also to identify any scale-free features, as many real-world networks exhibit these characteristics.

Small-world networks are characterized by two primary properties: a high clustering coefficient and a small mean-shortest path length. These networks often contain cliques or near-cliques,

which are sub-networks where nearly all nodes are interconnected. This high clustering coefficient signifies that nodes within these cliques are highly likely to be connected to one another. Additionally, in a small-world network, most pairs of nodes are connected by short paths, indicating a small average shortest path length between nodes.

Another prominent feature of small-world networks is the presence of hubs, or nodes with a high number of connections. These hubs facilitate short paths between other nodes in the network. (Wikipedia Contributors, 2024)

The degree distribution in small-world networks often follows a fat-tailed distribution, meaning that there is a higher than expected number of nodes with a high degree of connections. This distribution indicates the prevalence of hubs within the network. In networks with a power-law degree distribution, the deletion of a random node usually does not significantly impact the network's mean-shortest path length or clustering coefficient, since the majority of shortest paths pass through these hubs.

Barabási and Albert (1999) proposed that many real-world networks exhibit a power-law degree distribution, characterized by a few nodes with very high degrees. They also introduced the "preferential attachment" model, suggesting that new nodes are more likely to connect to existing nodes with a higher degree, thus creating a scale-free network with a power-law degree distribution. (Dai Shizuka, 2018)

In summary, small-world networks are marked by their high clustering, short average path lengths, and the significant role of hubs, all of which contribute to their unique structural properties. By examining these properties in our network, we aim to better understand its underlying structure and how it compares to other real-world networks.

Used software

For all computations and visualizations in this research, we utilized Python within the Jupyter Notebook environment. This choice allowed for an interactive and flexible workflow, which was essential for our complex analysis.

We employed several key Python packages to facilitate our work:

- **NetworkX**: This package was crucial for our social network analysis. It provided all the necessary measurements and data structures for representing and manipulating the graph.
- **Matplotlib.pyplot**: We used this package extensively for creating plots, graphs, and visualizing the network. Its versatility enabled us to present our data in a clear and comprehensive manner.
- **Concurrent.futures**: This package was instrumental for multithreading computations, particularly when calculating centrality values. It allowed us to optimize our computational efficiency.
- **Community**: This package was used for community detection within the network. It helped us identify clusters of nodes that are more densely connected internally than with the rest of the network.

By leveraging these tools, we were able to perform a thorough and detailed analysis of our social network dataset.

Measured values

Our analysis involved a comprehensive set of descriptive metrics to understand the structure and properties of the network. These metrics included:

- **Number of Nodes and Edges**: We quantified the total number of nodes (users) and edges (connections between users) in the network.
- **Density**: We measured the network density, which indicates the proportion of potential connections in the network that are actual connections.
- **Diameter**: We calculated the diameter of the network, representing the longest shortest path between any two nodes.

- Average Shortest Path: We determined the average shortest path length, which is the average distance between all pairs of nodes.
- Clustering Coefficient: We evaluated the clustering coefficient to assess the degree to which nodes in the network tend to cluster together.
- Number of Components and Connected Components: We analyzed the network's connectivity by identifying the number of components and specifically focusing on the largest connected component.

We also examined the degree distribution to understand the spread of connections among nodes.

For comparative analysis, we compared our network with several theoretical models:

- Erdős–Rényi (ER) Model: We compared our network with random graphs generated by the ER model.
- Watts-Strogatz Model: We used the Watts-Strogatz model to compare small-world properties.
- Barabási-Albert (BA) Model: We compared our network's properties with those of the BA model, known for its scale-free characteristics.

Additionally, we computed centrality measures to identify the most important nodes in the network:

- Degree Centrality: We measured the number of connections each node has.
- Betweenness Centrality: We assessed the extent to which nodes lie on the shortest paths between other nodes.
- PageRank: We used the PageRank algorithm to evaluate the relative importance of nodes.

Lastly, we conducted a community analysis to detect clusters within the network. This analysis helped us identify groups of nodes that are more densely connected internally than with the rest of the network.

We did not utilize measures such as Reciprocity and Transitivity because our network is undirected. However, it is worth noting that high transitivity is a characteristic often associated with small-world networks (Small World Networks - Math Insight, 2020).

Furthermore, we opted not to conduct Resilience Analysis, Diffusion over a network, Ego Network Analysis, and Homophily due to time constraints. While these analyses could provide valuable insights into the network, they did not directly contribute to our primary goal of demonstrating that this network exhibits small-world properties. These analyses could have shed light on aspects such as the network's reliance on central hubs for connectivity and the efficiency of information dissemination within the network.

Experimentation and Results

Sampled Twitch Social Network



Figure 2. Visualization of Twitch social network showing small world network features

Figure 5 illustrates concentrated clusters surrounding hub nodes, with sparse connections extending to nodes that have only a few connections or are minimally connected. This pattern indicates the characteristic structure of a small-world network.

Network descriptive statistics

After preprocessing our data, the network consists of **999 nodes** and **1550 edges**, significantly smaller than the original 168115-node graph, as previously explained.

The network's **density** is calculated at **0.0031**, suggesting sparse connections yet still indicating a highly interconnected graph. This characteristic aligns with our interest in small-world networks, where nodes maintain short average distances between them without being excessively interconnected.

The **diameter** of our network is **14**, which is notably compact considering the network comprises 999 nodes. This suggests that any node within the network can be reached from any other node in a relatively small number of steps.

The **average shortest path** in our network measures **4.7898**, a characteristic indicative of a small-world network. This concept aligns with Stanley Milgram's famous study in 1967 on "Six Degrees of Separation," suggesting that all nodes in the network can be interconnected within fewer than 5 steps.

The **cluster coefficient**, measured at **0.0529**, indicates a low tendency for nodes to form tightly-knit groups or clusters within the network. This observation suggests that the network may not fully exhibit small-world characteristics, as small-world networks typically feature higher cluster coefficients, fostering more interconnected local clusters despite their short average path lengths.

After conducting **connected components analysis**, we found that there is only **one component comprising all 999 nodes**. This number is likely influenced by the significant reduction in nodes during the sampling process. The presence of a single component indicates that the network is fully connected, without any isolated sub-networks. This connectivity ensures that every node in the network can reach any other node, either directly or through intermediate nodes. Therefore, the largest giant component identified through random selection encompasses the entire network.

Degree distribution

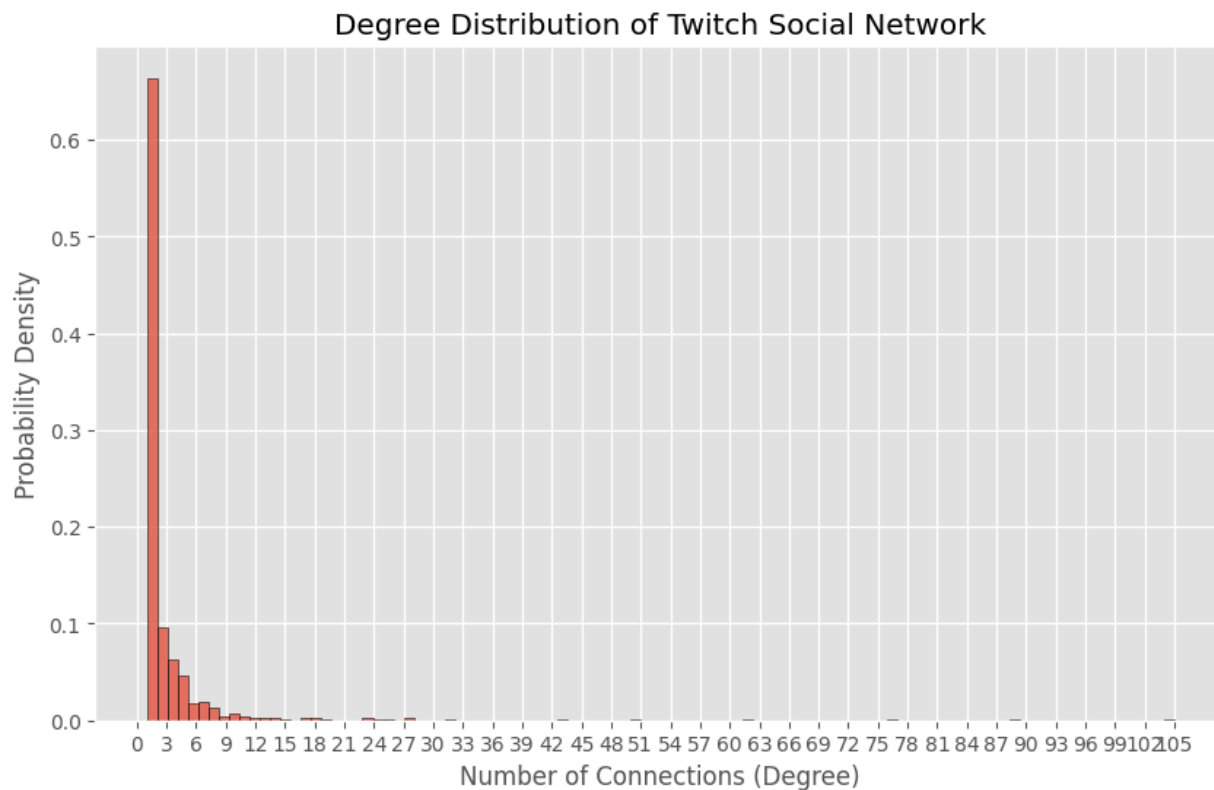


Figure 3. Degree distribution of Twitch social network

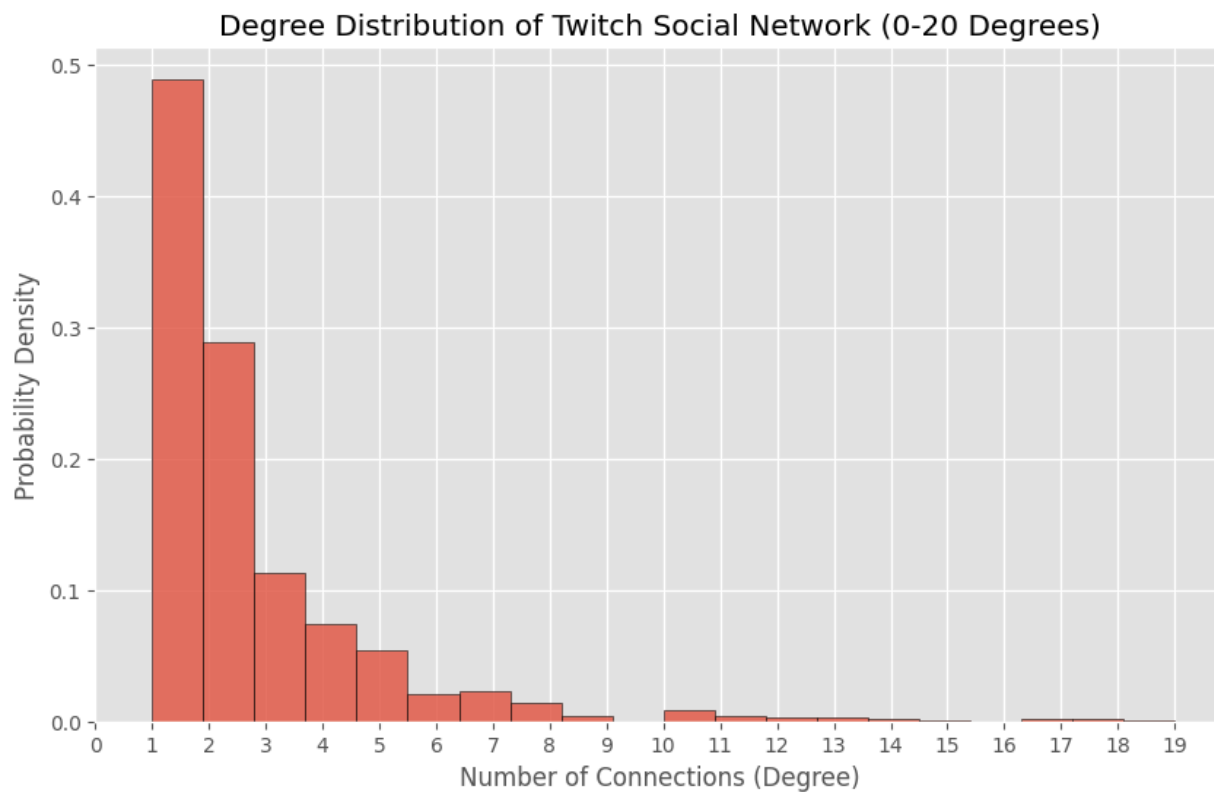


Figure 4. Degree distribution of Twitch social network

From figures 6 and 7, it is evident that the majority of nodes have a degree between 1 and 7, with 1 being the most common degree value observed. However, there are also nodes with very high degrees, some exceeding 100.

In real-world networks, it is typical for most nodes to have relatively low degrees, while a small number of nodes, known as hubs, have very high degrees, connecting them to numerous other nodes (The Degree Distribution of a Network - Math Insight, 2020).

The degree distribution data exhibits a fat-tailed distribution, indicative of scale-free network properties (The Degree Distribution of a Network - Math Insight, 2020). Moreover, the power law exponent of 3.03 derived from your network suggests that it follows a power law distribution.

Comparison to random models

Metric	Our Model	Random Graph (ER Model)	Small World (Watts-Strogatz) Model	Barabasi-Albert Model
Cluster Coefficient	0.0529	0.0014	0.4929	0.0440
Avg. Shortest Path	4.7898	Avg. shortest path: 6.1025 of LCC (not connected)	4.4385	2.9590

Figure 5. Table of comparison to random models

From Table 5, it is evident that the clustering coefficient in our network exceeds that of a random graph but falls significantly short of the small-world model, closely resembling the Barabasi-Albert model. This suggests characteristics typical of a scale-free network, with a clustering coefficient insufficient for a small-world designation.

Regarding the average shortest path, our network exhibits a shorter path length than a random graph, nearly matching the small-world model and surpassing the Barabasi-Albert model. This indicates prominent small-world features but a lack of pronounced scale-free characteristics.

Overall, while our network demonstrates higher clustering akin to scale-free networks and efficient connectivity resembling small-world networks, it does not align perfectly with either model individually.

Centrality measures

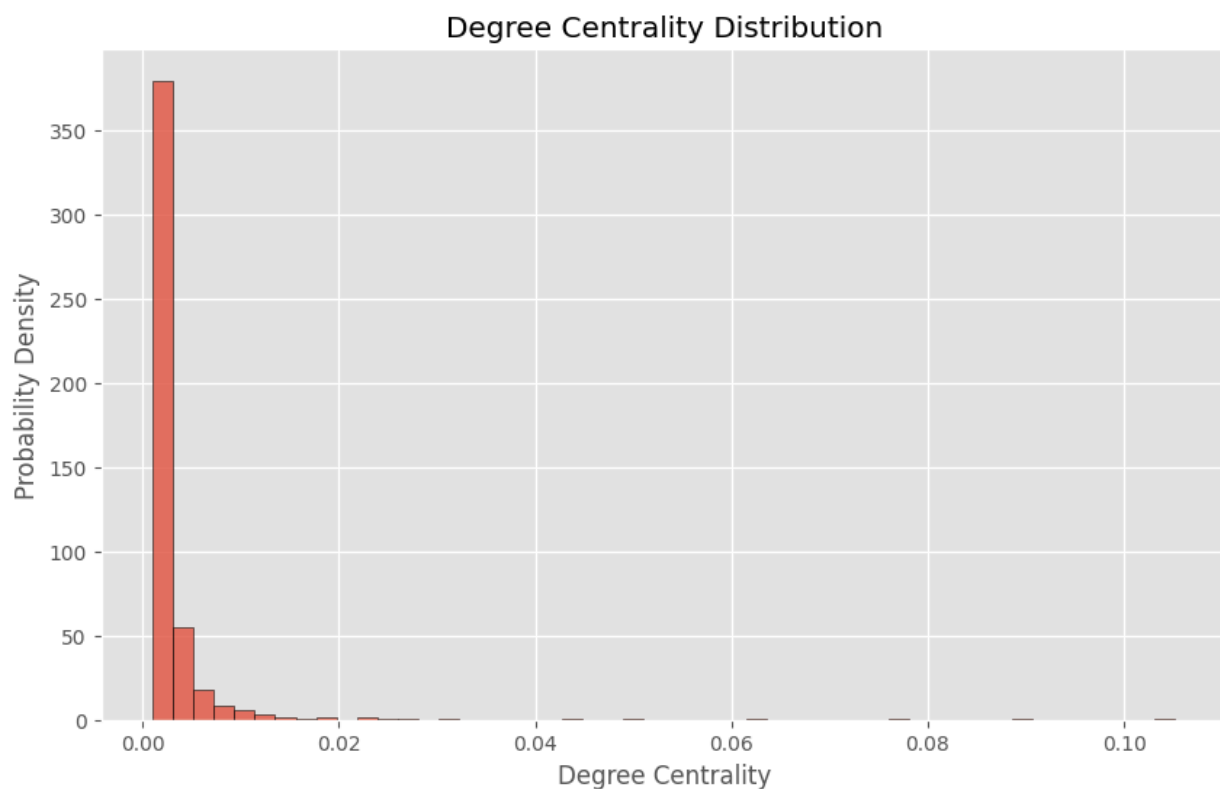


Figure 6. Plot of degree centrality distribution

Observing the figure 6, it is apparent that the majority of nodes exhibit low degree centrality, while a minority of nodes serve as hubs with high degree centrality. These hubs play a crucial role in enhancing network interconnectivity and facilitating efficient communication among nodes.

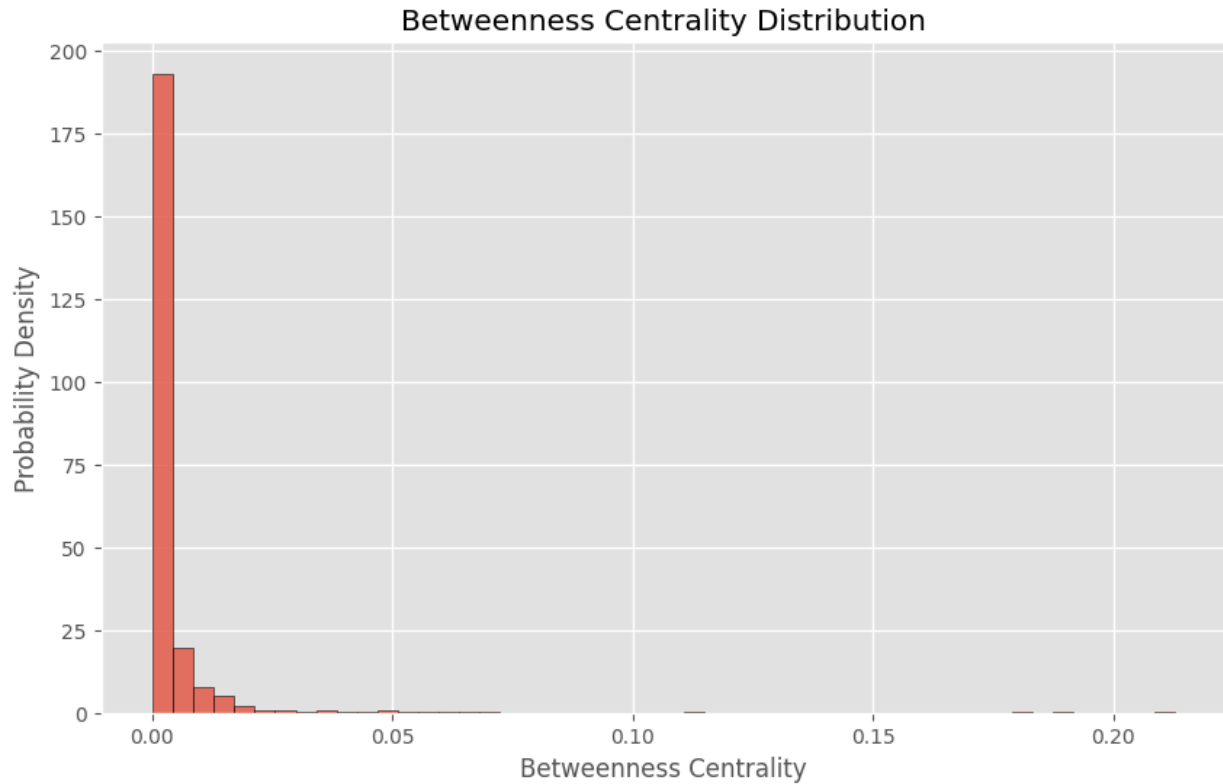


Figure 7. Plot of degree betweenness distribution

Nodes with high betweenness centrality serve as crucial bridges within the network, facilitating efficient communication paths. In small-world networks, these nodes play a pivotal role in maintaining short average path lengths and accelerating information dissemination. The figure 7 illustrates that while most nodes exhibit low betweenness centrality, a subset of nodes acts as significant hubs with high betweenness centrality, enhancing network connectivity.

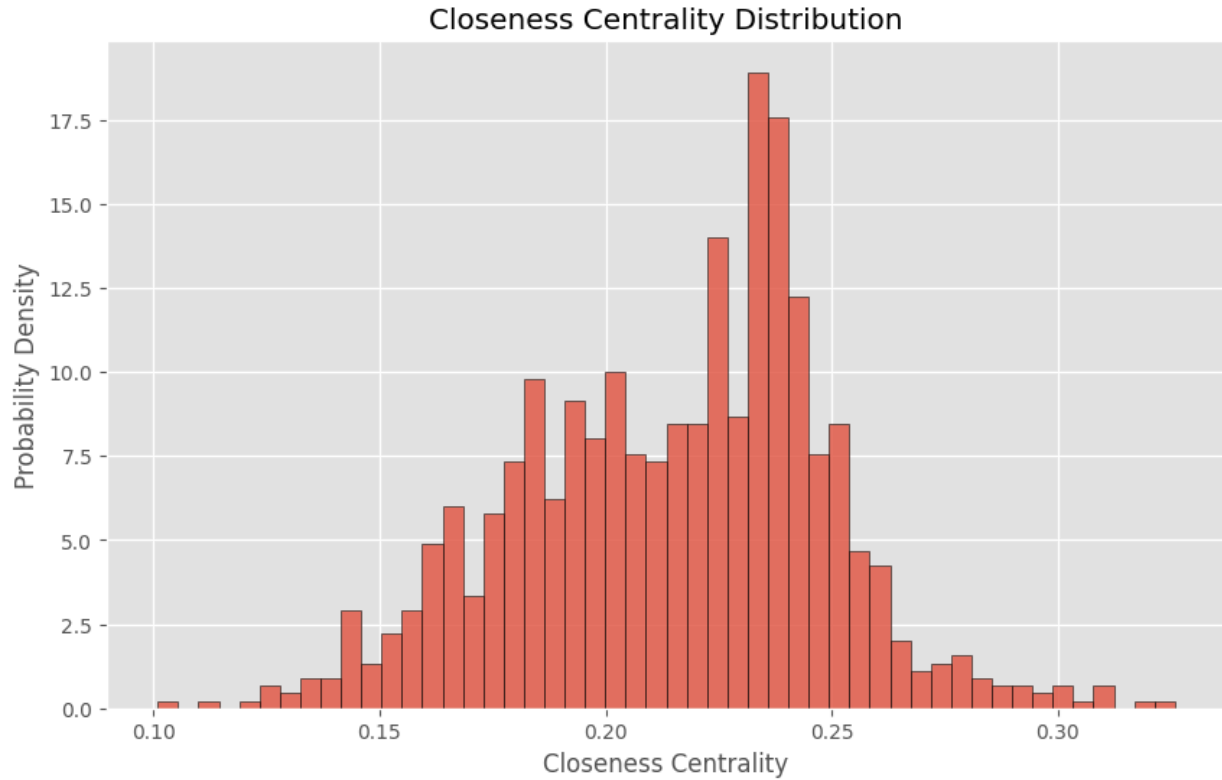


Figure 8. Plot of degree closeness distribution

The figure 8 depicts a Gaussian-like distribution of closeness centrality values, indicating that most nodes exhibit relatively high closeness centrality around 0.27. This distribution highlights the network's efficient connectivity, where nodes have short average paths to all other nodes.

This observation reinforces the small-world nature of the network.

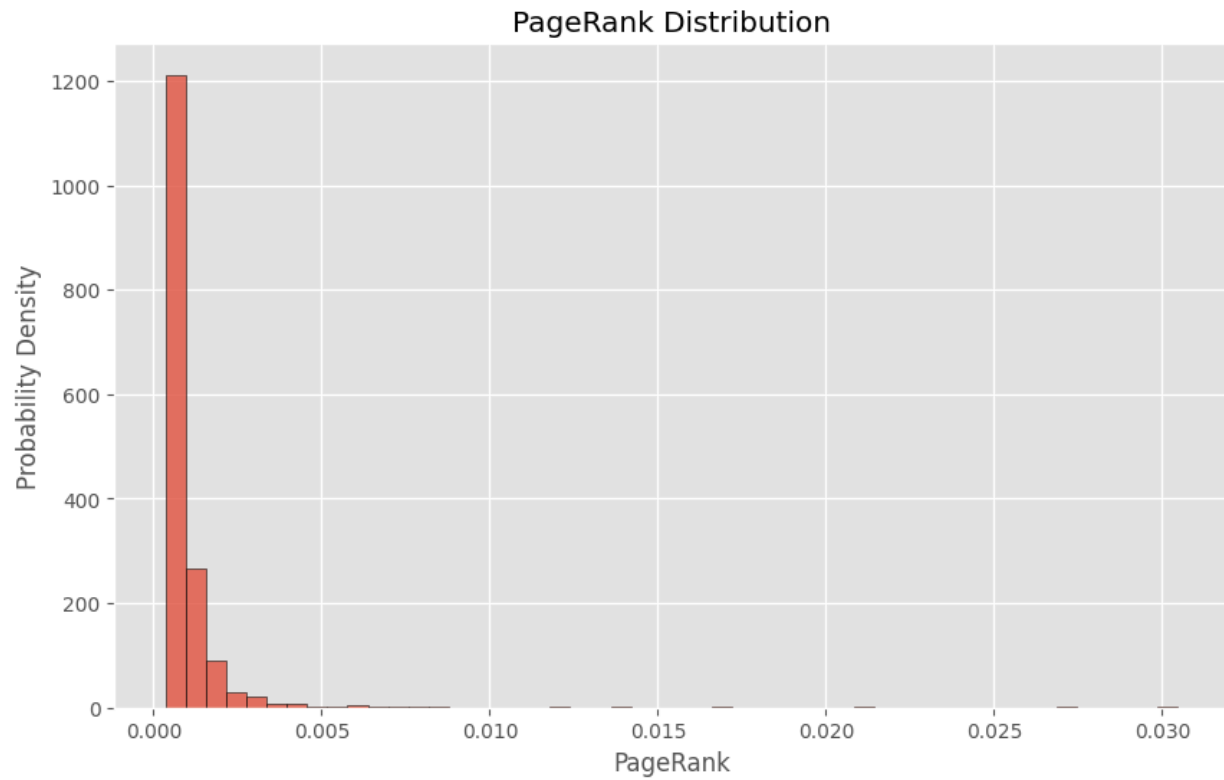


Figure 9. Plot of PageRank

The distribution observed resembles that of degree and betweenness centrality, indicating that the majority of nodes are not connected to influential nodes, while a subset of nodes holds significant influence within the network.

Community detection

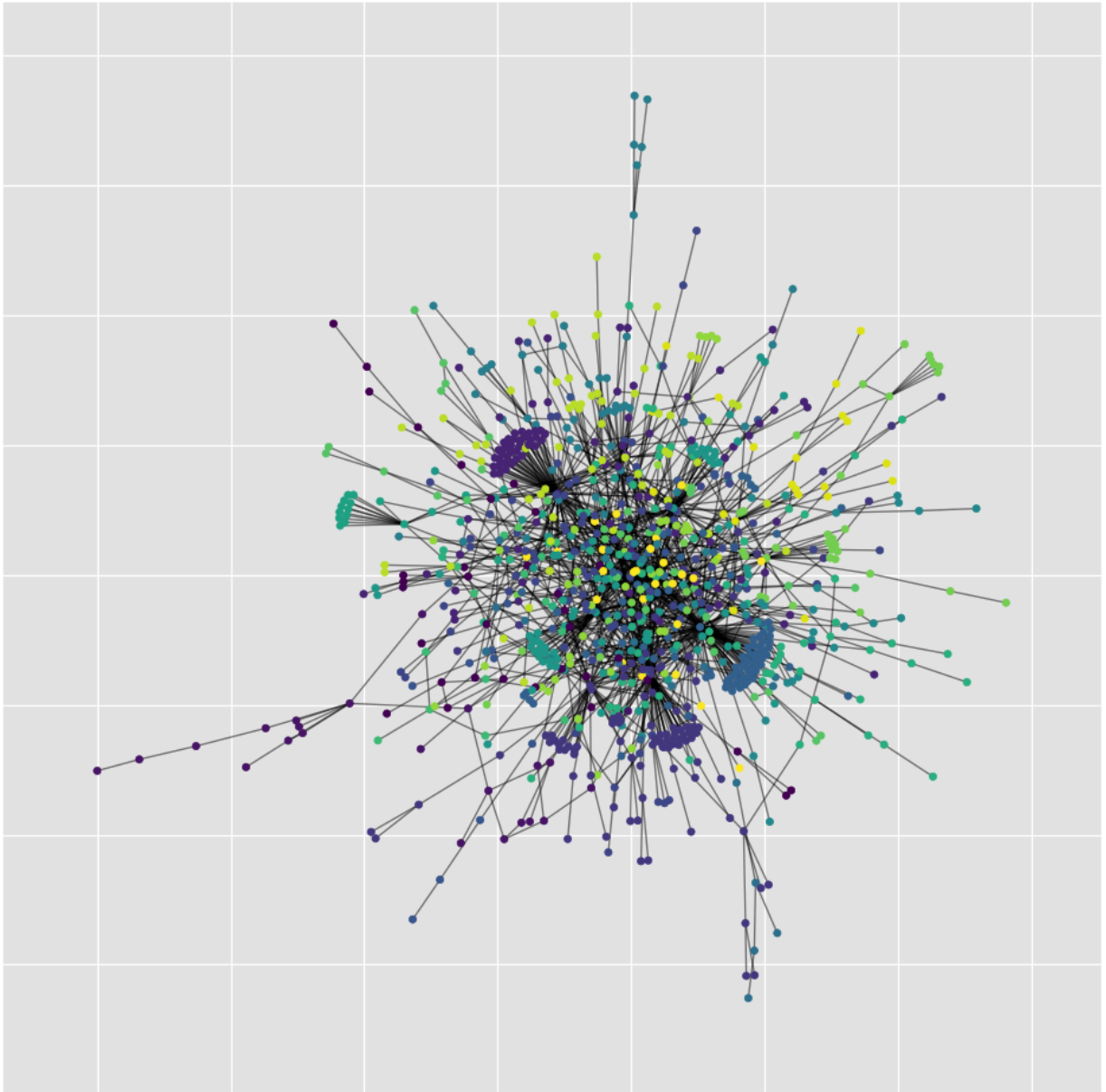


Figure 10. Visualization of the network with community detection

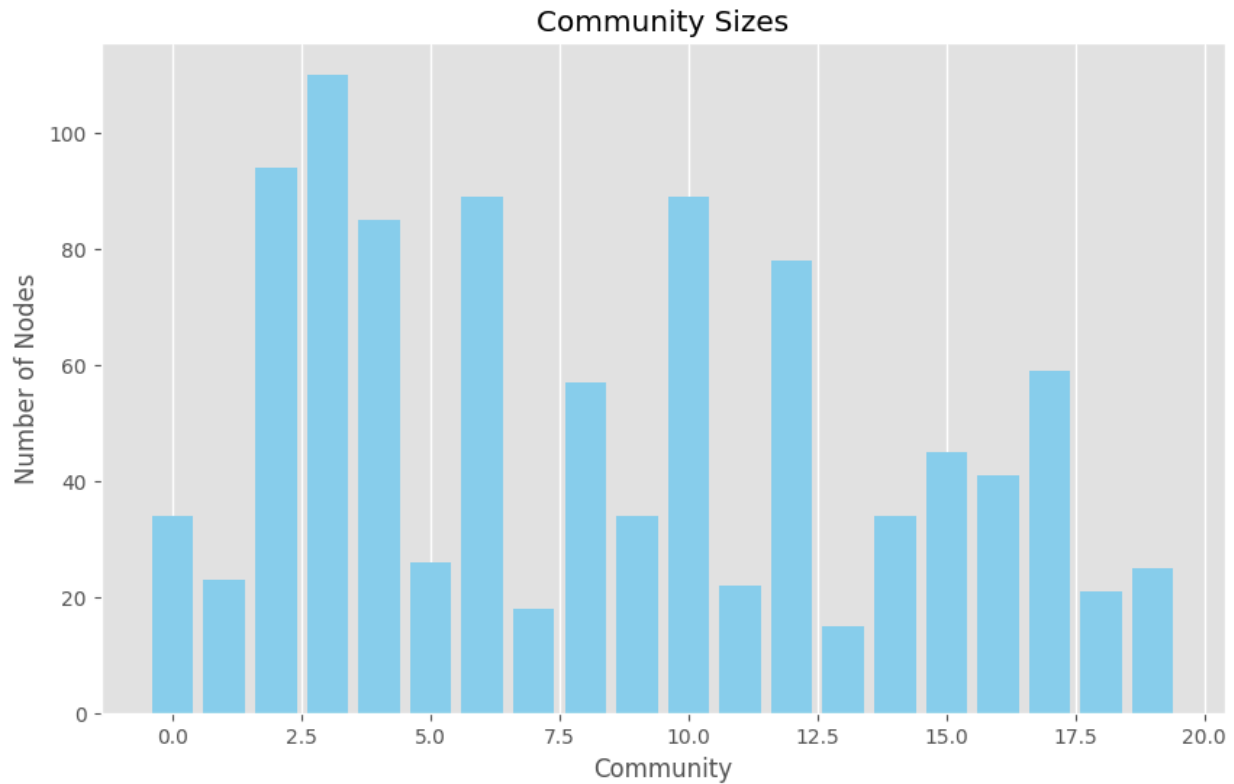


Figure 11. Community sizes

There are 20 distinct communities within the network. Some of these communities are quite large, with over 100 nodes, while others are smaller, consisting of around 20 nodes. This distribution demonstrates that the network forms both large communities with many nodes and smaller ones, all interconnected within the main network structure.

Discussion

In the work above, we notice that the clustering coefficient of the model is 0.0529, and it is very high even in comparison with 0.0014 of clustering coefficient in the random graph. A random graph will have nodes randomly connected leading to less likelihood of tight clusters.

This points out that the underlying structure of networks consists more or less of localized clusters. Networks probably have a higher degree of structure than random graphs. This higher clustering coefficient tells that these network models capture some of the properties of the observed network instead of putting together random links.

Even though we did not show the average shortest path of the random graph, the average shortest path distance of a random graph is usually less due to the fact that the nodes are randomly connected with each other. Our model exhibits an average shortest path of 4.7898, which is characteristic of small-world networks, as nodes can often be physically close to one another in a few steps. The model shows the same pattern as a small-world network when it comes to an average path length, but clustering coefficient is lower.

The average shortest path of the Barabasi-Albert model is 2.9590, which is a lot shorter than our model. This is a feature of scale-free graphs where a few and well-connected hub nodes impose a dramatic reduction in the average distances. The increase in distance among nodes indicates that the network is less centered, as compared to the Barabasi-Albert network. This elongated path may be a reason why the model seems to be more decentralized in terms of structure, in contrast to the centralized network.

A major limitation of this study is that the study uses the sole data. Under such constraints, the study is not sure of its generalizability. Different datasets bear unique structural properties and dynamic patterns, hence you may not fully detect the performance pattern of network MODEL without comparing different datasets. Therefore, future research should be conducted with more datasets for broader social generalization.

Conclusions

The study demonstrates that the network of Twitch users exhibits characteristics of a small-world network, as evidenced by the average shortest path length when compared to random models, centrality measures, and community detection results. However, the clustering coefficient indicates that the network also shares attributes with scale-free networks, similar to the Barabasi-Albert model. This suggests that the Twitch user network is a hybrid, possessing features of both small-world and scale-free networks. The network's structure supports efficient connectivity and rapid information dissemination through influential hub nodes.

In future research, it is essential to extend the study to more networks from various data sources related to Twitch to ensure the reliability and generalizability of the results. Additionally, incorporating features from the dataset will provide deeper insights into the network's behavior and underlying patterns. Utilizing greater computational power will enable the analysis of the entire network, rather than just a subset, offering a more comprehensive understanding. Furthermore, conducting a homophily analysis of features, such as language and content type, will help in understanding community building within the network. This multi-faceted approach will enrich the analysis and contribute to a more nuanced understanding of Twitch's social network dynamics.

References

1. *SNAP: Network datasets: Twitch Gamers Social Network*. (n.d.). Stanford Network Analysis Project. Retrieved June 23, 2024, from https://snap.stanford.edu/data/twitch_gamers.html
2. Wang, T., Chen, Y., Zhang, Z., Xu, T., Jin, L., Hui, P., Deng, B., & Li, X. (2011). Understanding Graph Sampling Algorithms for Social Network Analysis. <https://doi.org/10.1109/icdcs.2011.34>
3. Wikipedia Contributors. (2023, November 1). Snowball sampling. Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Snowball_sampling
4. Wikipedia Contributors. (2024, June 20). Giant component. Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Giant_component
5. Wikipedia Contributors. (2024, May 15). Small-world network. Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Small-world_network#Properties_of_small-world_networks

6. Dai Shizuka. (2018). Small World and Scale-Free Networks. Github.io.
https://dshizuka.github.io/networkanalysis/example1_smallworld.html
7. Small world networks - Math Insight. (2020). Mathinsight.org.
https://mathinsight.org/small_world_network
8. The degree distribution of a network - Math Insight. (2020). Mathinsight.org.
https://mathinsight.org/degree_distribution
9. Wikipedia Contributors. (2024, April 3). Fat-tailed distribution. Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Fat-tailed_distribution
10. Wikipedia Contributors. (2024, March 26). Scale-free network. Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Scale-free_network
11. Wikipedia Contributors. (2024, March 26). Scale-free network. Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Scale-free_network
12. Derr, A. (2023, September 14). *Social Network Analysis 101: Ultimate Guide - Visible Network Labs*. Visible Network Labs.
<https://visiblenetworklabs.com/guides/social-network-analysis-101/>